Unit 5

Applications, Use Cases, and the Quantum Future

Real-world application domains:

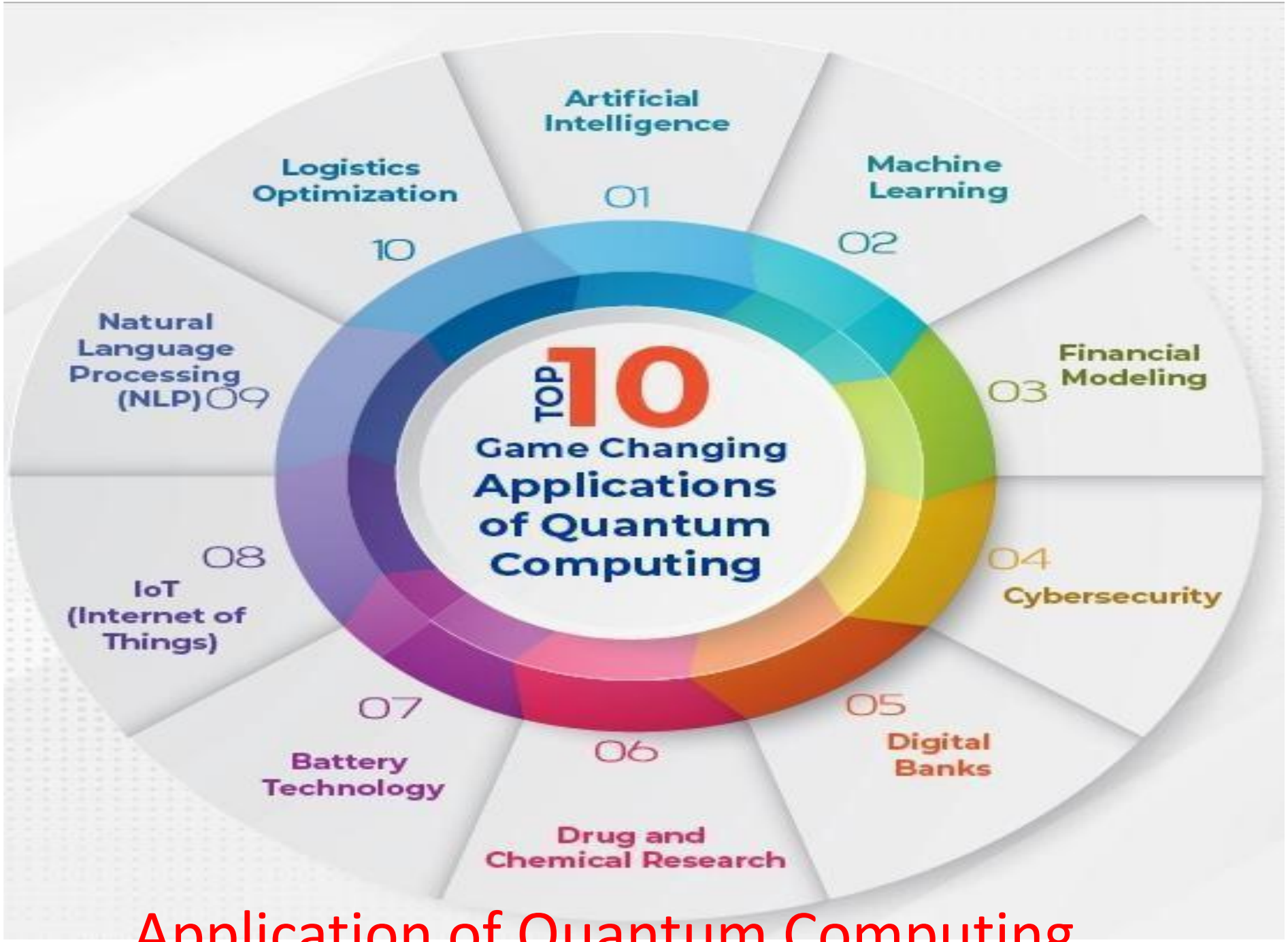
Healthcare (drug discovery), Material science, Logistics and optimization, Quantum sensing and precision timing, Industrial case studies: IBM, Google, Microsoft, PsiQuantum, Ethical, societal, and policy considerations, Challenges to adoption: cost, skills, standardization, Emerging careers in quantum: roles, skillsets, and preparation pathways, Educational and research landscape – India's opportunity in the global quantum race.

- Quantum computing promises to revolutionize diverse fields by addressing problems beyond classical capabilities.
- In medicine, it enables atomic-level simulations for drug discovery and personalized treatments.
- Finance benefits from enhanced risk analysis, portfolio optimization, and fraud detection.
- Logistics companies like DHL and Volkswagen use quantum algorithms to optimize routes and lower costs.
- Cybersecurity faces new challenges as current encryptions may be broken, leading to quantum-safe cryptography.

- In AI, quantum computing accelerates machine learning for autonomous driving and language translation.
- It improves climate modeling, material discovery for batteries and superconductors, and the simulation of quantum systems themselves.
- As this technology advances, it demands new programming paradigms, infrastructure, and ethical frameworks to harness its transformative potential responsibly.

5.1 Real-world application domains

- Quantum technologies are rapidly transitioning from theory to practical use, leveraging principles like superposition, entanglement, and tunneling to achieve tasks beyond the capabilities of classical systems.



Application of Quantum Computing

Real-world applications of quantum technologies, categorized by field:

1. Healthcare
2. **Material Science**
3. **Logistics and optimization**
4. **Quantum sensing and precision timing**

1. Healthcare

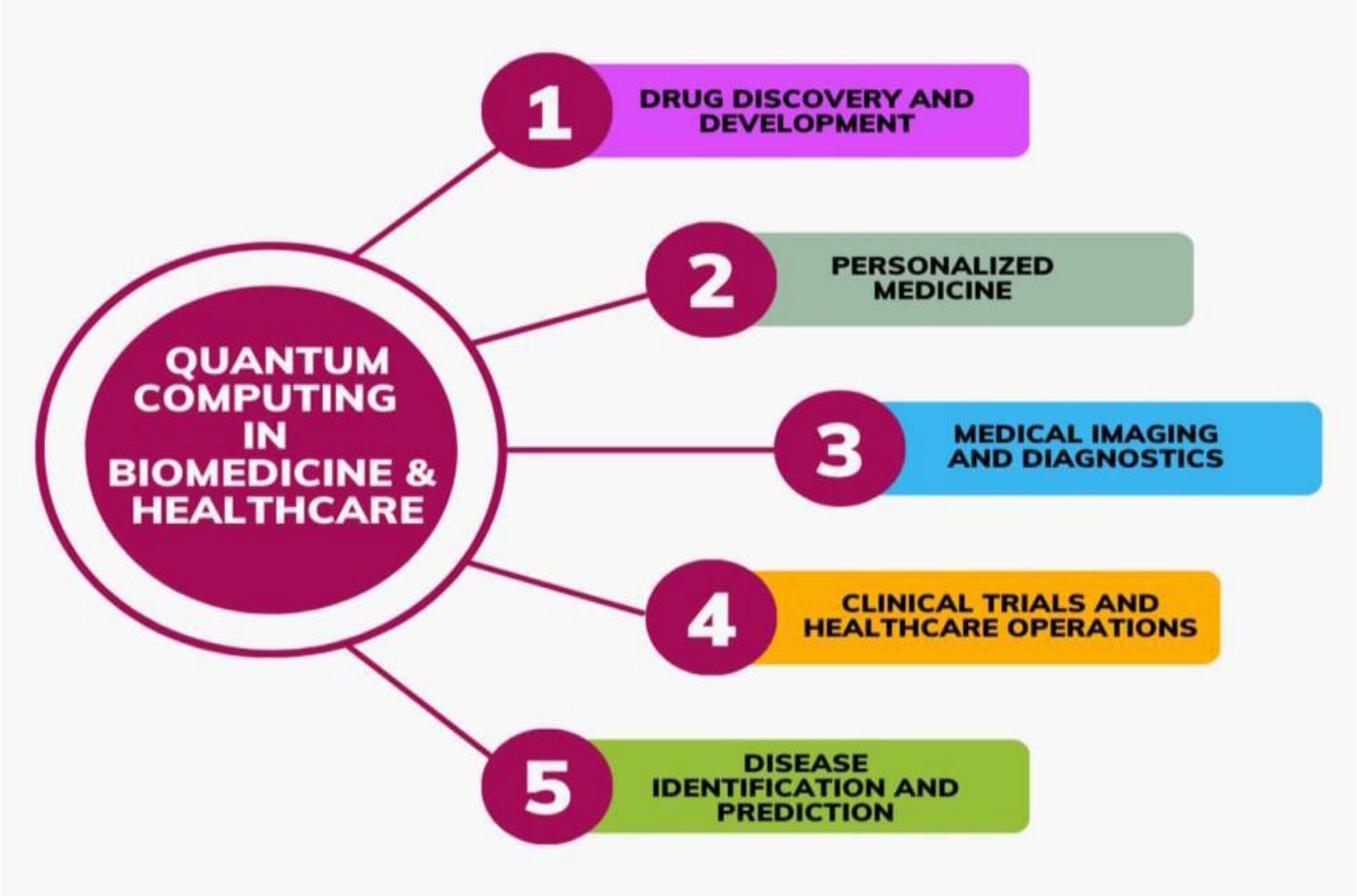
- Drug Discovery
- **Quantum Computing for Drug Discovery**
- **Current Applications & Progress**

Drug Discovery

- Drug discovery is one of the most promising real-world applications of quantum technologies.
- The process of discovering new drugs involves simulating complex molecules and chemical reactions—tasks that are extremely difficult and time-consuming for classical computers.
- Quantum computers offer a revolutionary approach

The five drug development phases

The Potential Role of Quantum Computing in Biomedicine and Healthcare



- Quantum computing is revolutionizing drug discovery by simulating complex molecular structures and chemical interactions far more efficiently than classical computers.
- Leading tech and biotech companies are using quantum algorithms to identify promising compounds, predict protein binding, and accelerate personalized medicine development.
- By leveraging quantum superposition and entanglement, quantum systems reduce the time and cost of finding new drugs, improve patient-specific treatments, and optimize clinical trials.
- These advances will help expedite responses to major diseases and pandemics in the future.

Quantum Computing for Drug Discovery

- **i. Molecular Simulation**
- **ii. Speed and Accuracy**
- **iii. Reduction in Cost & Time**

i. Molecular Simulation

- Molecules follow the laws of quantum mechanics. Quantum computers can naturally model these behaviors:
- Simulate interactions between atoms and molecules.
- Predict molecular properties and binding affinities.
- Understand reaction mechanisms at the quantum level.

ii. Speed and Accuracy

- Traditional supercomputers use approximations for quantum behavior, which limits accuracy. Quantum computers can perform these simulations exponentially faster and more accurately, leading to:
- • Faster screening of drug candidates.
- • Better prediction of side effects and efficacy.

iii. Reduction in Cost & Time

- Traditional drug development takes 10–15 years and billions of dollars. Quantum-enabled simulations could significantly shorten R&D cycles.

Current Applications & Progress

- 1. Protein Folding & Target Interaction
- Quantum computers help simulate how proteins fold and how drugs bind to them.
- Understanding folding is critical for targeting diseases like Alzheimer's, cancer, and viral infections.
- 2. Chemical Reaction Simulation
- Modeling how a candidate drug behaves in the human body.
- Example: BASF and Zapata Computing work on reaction pathway predictions.

Table 5.1 Companies and Research Labs Involved

Organization	**Contribution**
IBM Quantum	Simulated small molecules like LiH and BeH_2. Collaborating with biotech firms.
Google Quantum AI	Simulated basic molecules using Sycamore quantum processor.
D-Wave	Exploring quantum annealing for molecule optimization
AstraZeneca	Collaborating with Quantinuum and Cambridge Quantum for drug design.
Roche & Boehringer Ingelheim	Partnering with quantum startups to simulate complex molecules.
ProteinQure	Uses quantum computers for protein-drug interactions and optimization

II. Material Science

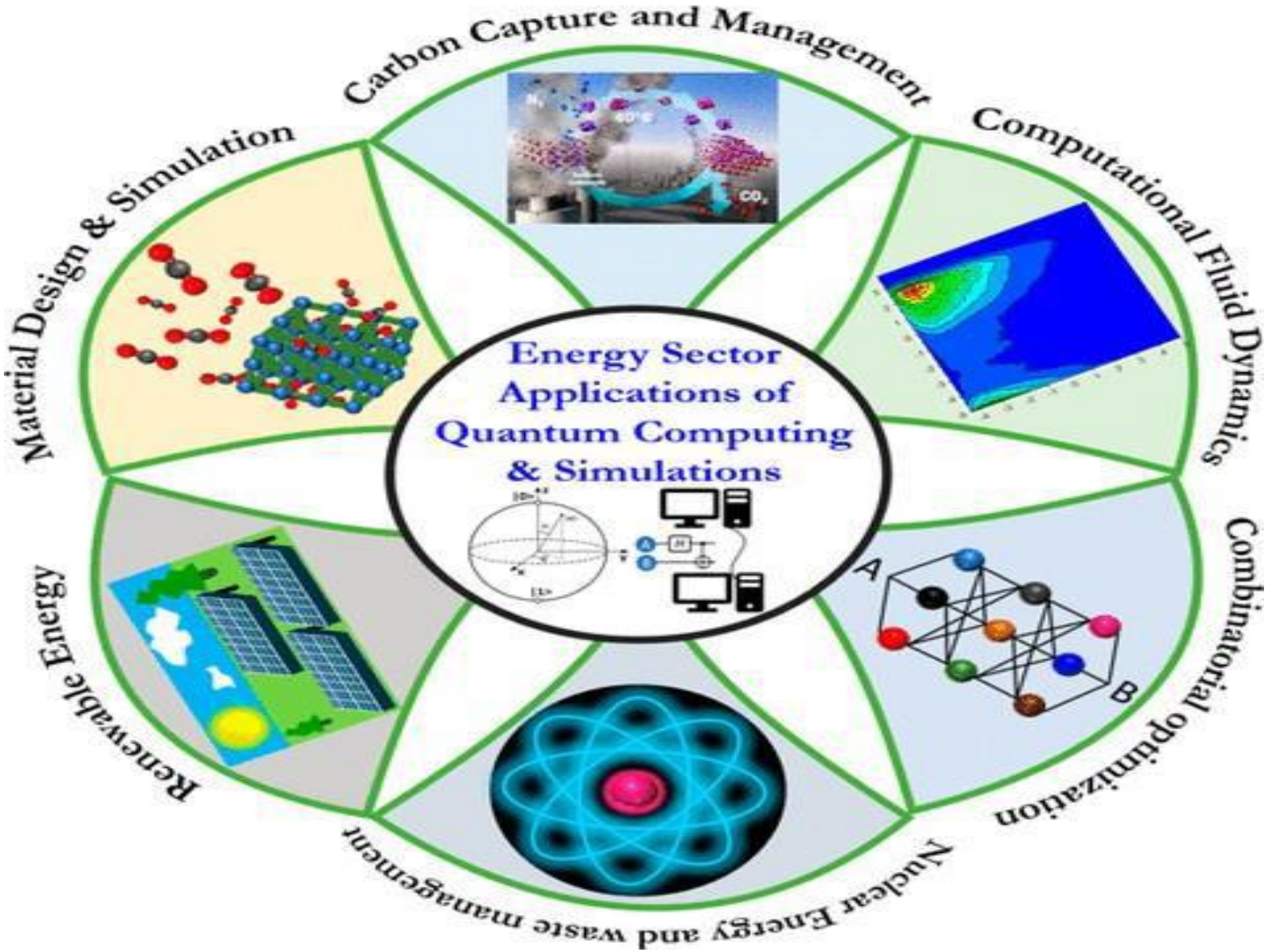
- Material science is an interdisciplinary field that studies the structure, properties, processing, and performance of materials to understand how they can be used and improved for various applications.
- Quantum technologies are transforming material science by enabling scientists to discover and design new materials with unprecedented accuracy and speed.
- Quantum computers and quantum simulations help model complex atomic interactions that are too difficult for classical computers to handle.

Why Use Quantum Technologies in Material Science?

- 1.Quantum systems obey quantum rules
- Traditional materials modeling often relies on approximations. Quantum computers simulate matter at the quantum level—electrons, bonds, energy states—without such approximations.
- 2.Designing from the atom up
- Quantum technologies allow researchers to:
- Discover new superconductors.
- Design stronger, lighter alloys.
- Engineer better batteries, semiconductors, and catalysts.

- Quantum computing enables precise simulation of material properties at the atomic scale, surpassing traditional systems and accelerating the discovery of advanced materials.
- This breakthrough supports the design of superior superconductors, lightweight alloys, advanced polymers, and battery materials tailored for energy, electronics, and aerospace industries.
- Researchers use quantum simulations to modify atomic structures virtually, optimizing materials before physical creation—thus saving significant time and resources.

Quantum Computing and Simulations for Energy Applications



- Quantum computing accelerates the development of advanced batteries and more efficient solar panels by enabling direct modeling of quantum effects in materials, providing a level of scale and precision beyond high-performance classical computing.
- This capability supports breakthroughs in sustainability, electronics, and manufacturing.
- For example, Volkswagen is actively using quantum simulation to innovate next-generation battery materials, aiming for safer, lighter, and more powerful batteries.

5.1.3 Logistics and optimization

- Quantum computing is set to revolutionize logistics and optimization problems that are computationally intensive for classical systems. These include route optimization, supply chain management, inventory forecasting, and delivery scheduling.
- Quantum algorithms like the Quantum Approximate Optimization Algorithm (QAOA) are being explored to solve such combinatorial problems more efficiently.

- Conventional computing is used to sync with operating models that need to continuously compute and recalculate optimal routes of traffic management, fleet activities, airport regulation, cargo and distribution, and that could seriously affect applications
- Some of these tasks are more complex for a classic computer but can perform quite efficiently with quantum computers
- Two common approaches to solve these issues are – Quantum Annealing and Universal Quantum
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Quantum Computers

Improved speed

Stores more information

Uses less energy

Advanced dynamic route optimization

Maximize simultaneous packing of parcels

Support adaptive reallocation of assets

Enable quick testing of designs and materials for logistic use

- Quantum computing is positioned to transform logistics and optimization by effectively tackling computationally intensive challenges like route optimization, supply chain management, inventory forecasting, and delivery scheduling.
- Quantum algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) and quantum annealing can solve complex combinatorial problems much faster than classical methods, handling large numbers of variables and constraints.
- Major logistics companies are already exploring these algorithms to optimize deliveries, reduce costs, and improve efficiency across supply chains and transportation networks

- Companies like DHL and FedEx are exploring quantum computing to reduce delivery times and costs under constraints such as traffic and weather.
- Quantum systems also optimize production workflows, resource allocation in manufacturing, and complex scheduling problems in industries like airlines.
- As logistics become more complex globally, quantum computing offers scalable solutions to NP-hard problems where classical systems fail, promising significant advantages in speed and precision for future commerce.

5.1.4 Quantum sensing and precision timing

- Quantum sensing utilizes the unique properties of quantum systems—such as superposition, entanglement, and tunneling—to detect and measure physical quantities with unprecedented precision and sensitivity, far beyond classical sensors.
- Quantum sensors can measure minute changes in magnetic, gravitational, and electric fields, as well as acceleration and rotation, overcoming limitations like thermal noise and environmental interference.
- This enables breakthroughs in fields needing ultra-high sensitivity, such as early earthquake detection, underground resource mapping, medical diagnostics, and advanced navigation.

- Precision timing, achieved through quantum clocks, delivers highly accurate synchronization vital for GPS, financial networks, and secure communications, surpassing current atomic clocks in stability and accuracy.
- Quantum gravimeters and magnetometers further support applications in geological surveying, oil and gas exploration, defense, and seismic monitoring, making quantum sensing crucial for next-generation imaging, surveying, and timing technologies.

- Quantum sensors are transforming medical diagnostics by enabling non-invasive, highly detailed monitoring of brain and heart activity through techniques like quantum-enhanced magnetoencephalography (MEG) and magnetocardiography (MCG).
- These technologies provide higher resolution than conventional imaging, allowing earlier detection of neurological and cardiac conditions.
- Precision timing, using advanced quantum clocks based on atoms like strontium or ytterbium, is now essential for ultra-accurate synchronization in GPS,

- In defense and aerospace, quantum sensors are used for inertial navigation without GPS, improving accuracy for submarines, aircraft, or spacecraft—crucial where satellite signals can't be relied upon. As these sensors become more compact and efficient, they are expected in consumer electronics such as smartphones and wearables, driving advances in personal health tracking and environmental monitoring.
- Overall, quantum sensing and precision timing will quietly but fundamentally redefine measurement limits and technology capabilities across healthcare, navigation, security, and everyday life.

5.2 Industrial Case Studies

5.2.1 IBM

- IBM is a leading pioneer in quantum computing through its IBM Quantum platform, which offers cloud-based access to quantum processors for researchers and developers.
- Its open-source Qiskit framework promotes quantum programming and collaboration. The IBM Quantum System One, the first integrated commercial quantum system, is deployed globally.
- IBM's roadmap targets scaling from hundreds to thousands of error-corrected qubits, with major milestones like the 433-qubit *Osprey* (2023) and the upcoming 1,000-plus-qubit *Condor* (2025).

- IBM is advancing quantum error correction with LDPC codes and developing modular quantum computing, linking smaller chips for greater scalability.
- The Qiskit Runtime environment enhances performance through improved compilation and error mitigation.
- Their annual quantum roadmap, extensive education initiatives, and collaborations with CERN, MIT, and industries such as healthcare, finance, and chemicals highlight IBM’s leadership in technology and ecosystem growth.
- IBM’s efforts unify hardware progress, software innovation, and global community building in the quantum era.

5.2.2 Google

- Google gained prominence in 2019 by achieving quantum supremacy with its 53-qubit *Sycamore* processor, completing a computation in seconds that would take classical supercomputers millennia. The company aims to build a fault-tolerant quantum computer with around 1 million qubits, focusing on applications in AI, optimization, and quantum chemistry.
- Its Quantum AI Campus in Santa Barbara leads research in cryogenics, qubit control, and error correction techniques like surface codes and zero-noise extrapolation. Google is advancing hybrid quantum-AI models, combining classical and quantum computing for machine learning acceleration.

- Through its open-source platforms Cirq and TensorFlow Quantum, Google supports global research in quantum machine learning and simulation. Future plans include integrating quantum technology into Google Cloud, enabling enterprise access.
- Collaborations with institutions like UC Berkeley and ETH Zurich strengthen its scientific impact. Guided by a long-term vision of building a commercial, error-corrected quantum computer, Google continues to drive innovation across the global quantum ecosystem.